
Privacy Analysis of Augmented Reality Glasses

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Weekly Report 6

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What has been done (Previous Week)

- Bought the Ubertooth one, which is another Bluetooth sniffing device that is able to sniff both classic and ble traffic.
- Successfully set up and sniffed traffic on the Ubertooth one (both classic and ble), this was just for trial and was not a full experiment.
- I technically also bought another sniffer that just got delivered today (Nov 2nd), but since the Ubertooth one works fine right now, I will try this one later, but will not spend too much effort on it. It is a Nordic nRF52840 sniffer, which is from the same company that made the Adafruit chip, but this one supports BLE 5.0. So it might be able to help me understand whether it could be an issue of the different Bluetooth version.
- Experienced issues with the pairing between the phone and glasses regarding importing media captures, so the settings would say that the phone is connected, as well as on the top left corner of the app. I would be able to use voice commands that would require connection to the phone, but if I tried to import the photos taken, I would get a pop up saying that the phone cannot connect to the glasses.
- Have tried the methods written previously on the documentation but was not successful. This time I needed to uninstall and re-install the app for it to import captures again.
- However, for the re-install of the app, this time they demanded an email account for logging in, which I was able to avoid the first time. I used a 10 minute mail, which is now also linked to the instagram account).

What will be done next week

- Capture the Bluetooth traffic for both classic and ble with the following:
 - Experiment is similar as when sniffing Wifi traffic, so having two different set of experiments where one is using voice commands, the other using touch commands, and a third for when unexpected issues occur during the experiments.
 - But also including another aspect where there will be another split for - With / Without Internet Connection.
 - These experiments also needs to include the initial pairing of the glasses to the phone, so the glasses will be unpaired from the phone in the beginning when the experiment begins.
 - Include the documentation for the experiments.

- And a voice recording of the experiments
- The focus of the week is to write/add documentation in regards to:
 - The Bluetooth experiments, including a basic description of the particular experiment, the metadata, the commands given, and some additional info (like some timestamps or if some problems occurred during the experiment).
 - The Wi-fi experiments, the main part is already written for those experiments, it is mainly the additional info that needs to be added.
 - The pairing issue I experienced to the document 'Problems encountered with the glasses'.
 - The different Bluetooth sniffers to the document 'Bluetooth sniffer issues', the micro:bit and adafruit already has documentation, but need to add what I tried the week before as well. Need to add a section for the Ubertooth one.
- Have a static analysis using the MobSF app (will need to pull the apk from the phone and install docker)
- Look into decrypting the Bluetooth traffic

Meeting

Had a meeting with Nima on the the 1st, where:

- Briefed through the static analysis of the app using MobSF, including how to pull the apk and how to install Docker
- Discussed the captures for Bluetooth and network capture and their decryption
- Discussed about rooting the phone